

PulsoKit

Kit editor for the Roland TD-17 drum module

User manual

version 1.0 · macOS 14 or newer
github.com/arkus1972/PulsoKit

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1. What PulsoKit is, and what you need

PulsoKit is a kit editor for the **Roland TD-17** drum module. It lets you change your kit settings from a computer — on a big screen, with a mouse — instead of clicking through small menus on the module itself.

It talks to the module over MIDI, the same way the module talks to your computer when you record. It installs no drivers and needs no internet connection.

What you need

- A Mac running **macOS 14** or newer
- A **Roland TD-17** connected to it with a USB cable
- The module switched on **before** you start the app

What it does

- edits every kit parameter: instruments, levels, pan, V-Edit, the SUB layer, sends, pad EQ and MIDI note assignments,
- lets you hear your changes *before* they are written permanently,
- switches the current kit on the module from the app,
- backs up a kit and restores individual settings from that backup,
- opens and saves `.TD0` kit files,
- shows the trigger settings of all ten inputs.

What it does not do

- it does not change trigger settings — it only displays them,
- it does not upload your own samples to the module,
- it does not record audio and does not work as an instrument inside a recording program.

Worth knowing

PulsoKit never connects to the internet, collects nothing, and reads no files other than the ones you point it at yourself. It also writes nothing to your disk except the files you explicitly ask it to save.

2. Installing and opening it the first time

1. Unzip the downloaded `PulsoKit.zip`.
2. Move **PulsoKit** into your **Applications** folder.
3. Open it.

The first time, macOS will refuse. That is normal and does not mean anything is wrong with the app — it simply is not registered with Apple, and registering costs 99 dollars a year.

How to let it through

1. Open **System Settings**
2. Go to **Privacy & Security**
3. Scroll all the way down — you will see “PulsoKit was blocked”
4. Click **Open Anyway**

Once is enough. It opens normally from then on.

Never paste commands from the internet into Terminal

If you ever come across advice along the lines of “the app says it is damaged, paste this command into Terminal and it will go away” — **don’t**. That kind of command switches off the system’s file checking, malware scanning included. The exact same instruction is shipped with counterfeit software, and it works precisely because honest authors sometimes publish it themselves.

The message **“the app is damaged”** almost always just means the download was incomplete. Download it again.

Connecting the module

Switch the TD-17 on **before** starting the app and connect it with a USB cable. On launch the app finds the module by itself and loads whichever kit is currently selected on it.

At the top of the window, in the connection bar, you should see a green dot and *TD-17 detected*, along with the chosen MIDI input and output. If you have several MIDI devices connected, the app will not pick one for you — choose the TD-17 from the drop-down lists yourself.

3. Before you change anything — make a backup

Read this before your first edit

PulsoKit writes **straight into your module’s memory**. Once a change is written there is no undo button for it — the module keeps no history and has no trash. The only way back to how something sounded before is a backup you made earlier.

A backup takes about fifteen seconds:

1. Select the kit you want to protect on the module
2. Open the **Backup** tab in the app
3. Click **Export** and save the file somewhere you will find it

You get a `.json` file holding every setting of that one kit. It is worth making one backup per kit you intend to work on, and keeping them together in their own folder.

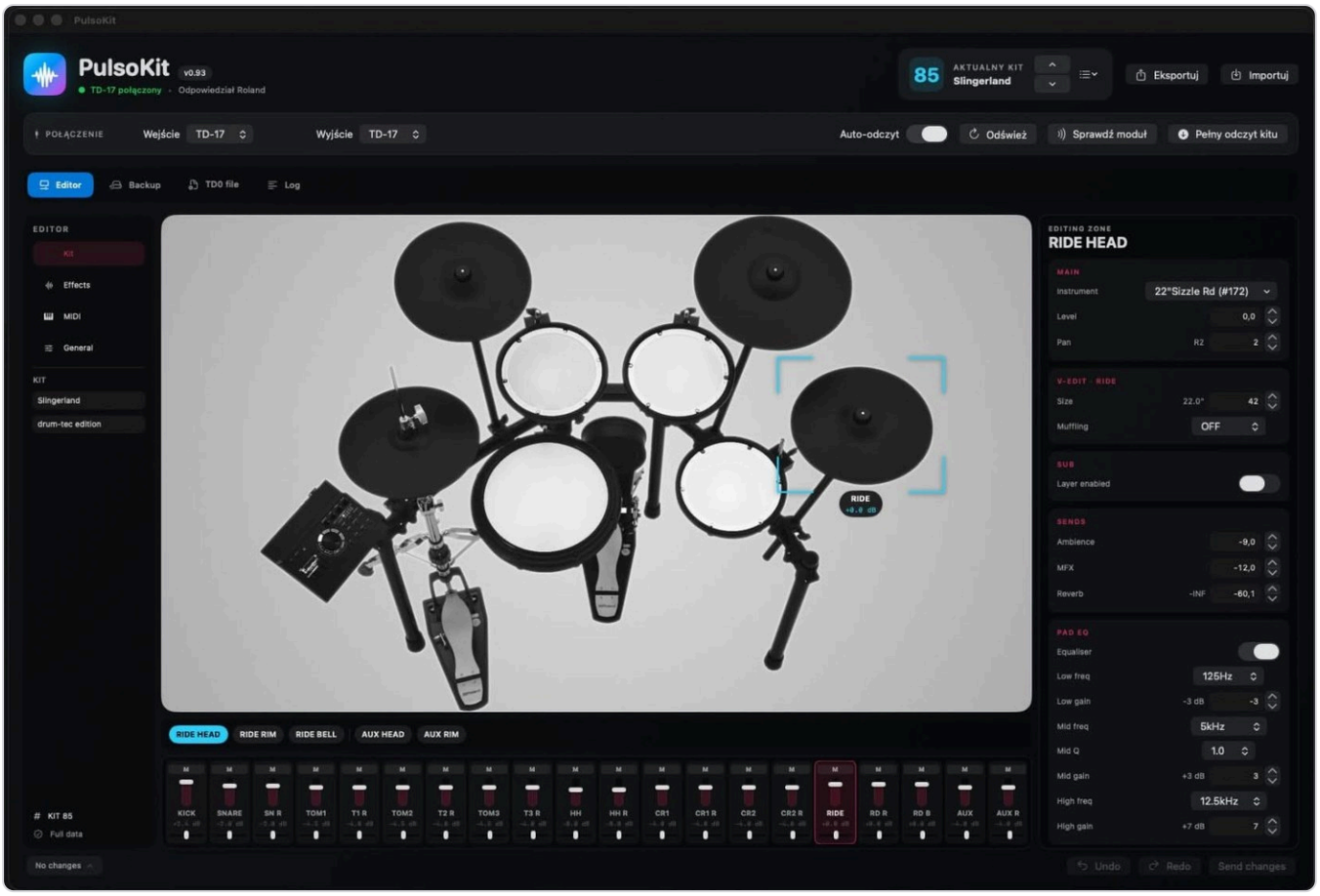
Especially before “Fast write of the whole kit”

That operation overwrites the **entire** kit at once and cannot be undone. Always export the kit first.

Good habits

- Don’t experiment on the kit you play live. Copy it to a free slot on the module first and work on the copy.
- Don’t unplug the cable or switch the module off mid-write. The app is writing whenever you can see a progress indicator.
- After a larger write the app asks for thirty seconds of quiet. Don’t change kits or power off during that time — the module is committing the data to its permanent memory.

4. The main window



The main window, **Editor** tab. This screenshot is from an earlier build and shows Polish labels — version 1.0 is in English, but the layout is identical.

ELEMENT	WHAT IT IS FOR
Connection bar (top)	MIDI input and output selection, module detection status, manual refresh and full kit read.
Current kit card	The number and name of the kit selected on the module. Clicking it opens a list of all one hundred kits — picking one switches the kit on the module too.
Section tabs	**Editor**, **Triggers**, **Backup**, **TD0 file**, **Log**.
Page bar (left)	Inside the editor: **Kit** (the pads), **Effects**, **MIDI**, **General**. Below it, the kit name and description.
The stage	A picture of the kit. Clicking a pad selects it for editing — and plays it, if auditioning is on.
Zone strip (below the stage)	A level fader for every zone plus an **M** button to mute it. Drag along the groove to change the level.
Inspector (right column)	Every parameter of the selected zone: instrument, level, pan, V-Edit, SUB layer, sends, pad EQ.

Bottom bar	Auditioning, the pending-changes counter, undo, try-before-write and send.

5. Editing a kit

There is one rule worth remembering above all others:

> **Nothing reaches the module on its own**
> No fader, no field and no drop-down writes anything to the module. Every edit goes onto a **pending list** and waits there until you deliberately press the send button. Experiment freely.

How to edit

1. Click a pad on the stage to select a zone.
2. If the pad has several zones (head, rim, bell), pick the right one from the chips below the stage.
3. Change parameters in the inspector on the right.

You can type numbers on the keyboard or use the arrows. A value outside the allowed range is clamped to the nearest one that fits.

The pending list

The bottom bar shows a counter — for example *3 pending changes*. Clicking it opens the list: each entry shows what the change affects, the old value and the new one. The cross next to an entry cancels that one change.

Undo

Undo and **Redo** work on the pending list — they step back and forth through your edits before anything goes to the module.

> **Changing kits discards unsent edits**
> If you switch kits while editing — in the app or with the knob on the module — the app loads the new kit and your unsent changes are discarded. It says so plainly in the bar at the bottom. Send your changes before switching kits.

6. Auditioning and trying changes first

The app has no sounds of its own — they all live in the module's memory. So it sends the TD-17 an ordinary MIDI note, and the sound comes out of the module exactly as if you had hit the pad with a stick.

Auditioning a single pad

When the speaker icon in the bottom bar is on, clicking a pad on the stage plays it. The slider next to it sets how hard. **Play kit** walks through the whole kit pad by pad, and turns into **Stop** while it runs.

Trying before writing — the "Try it" button

This answers the question "will this actually sound good?" *before* you make it permanent. Because the sounds live in the module, the only way to hear anything is to put your edits into the module — so the app writes them **on trial**, remembering exactly what was there before.

1. Make the changes you want to check.
2. Press **Try it**. The app sends them to the module and immediately plays the pads they affect.
3. Listen. Now you have two ways out:

BUTTON	WHAT IT DOES
Undo in module	Puts the pre-trial values back into the module. Your edits stay on the pending list, so you can keep working on them.
Keep	Keeps what you are hearing. The changes become permanent.

Changing an instrument is the exception

Writing a different instrument wipes that pad's V-Edit settings in the module and replaces them with the factory ones for the new sound. Going back restores the old instrument, but its V-Edit will be the factory version, not yours. The app warns you with an orange notice before you press.

Everything else — levels, pan, sends, EQ, notes — comes back without a trace.

7. Sending changes to the module

The **Send** button in the bottom right writes all pending changes. The app sends them one at a time and, after each one, asks the module what it actually stored — a change only leaves the list once it is confirmed.

While it runs you see the progress (“2 of 7”), and the button turns into **Stop** so you can interrupt it.

What the messages mean

MESSAGE	MEANING
Module did not confirm…	The module did not confirm that value — it may have rejected it or clamped it to its own range. The rest of the batch was held back. Whatever did not go through stays on the list.
Send stopped…	You interrupted the send. Anything already written is in the module.
Kit reloaded — … discarded	The kit was reloaded (you changed it on the module, for instance) and unsent changes were discarded, because they applied to the previous state.
Settling in the module (30 s)	The module is committing data to permanent memory. Don’t change kits or switch it off during this time.

Changing kits from the app

Click the current kit card at the top — a list of all one hundred kits opens, with their names. Picking one switches the kit on the module as well, and the app loads its contents straight away.

8. Backups

The **Backup** tab handles backups of a single kit.

Export

Saves the current kit to a `.json` file. The file is readable text — you can open it in any editor and see what is inside. One condition: the kit must have been fully read first.

Import and compare

Once a file is loaded, the app compares it with the kit currently in the module and lists every difference. Importing on its own writes nothing — it is only a preview.

Restoring

You have two routes:

- **One value at a time** — each difference has its own restore button, with a confirmation and a read-back check. Safe, and usually all you need.
- **Fast write of the whole kit** — overwrites the entire kit with the backup's contents in a few dozen blocks, then reads everything back and compares.

The fast write overwrites the whole kit

It cannot be undone. Export the target kit before you run it, and you will always have something to go back to.

If the cable comes out or the module switches off partway through, the app stops and tells you exactly which block it stopped at and that the kit is now incomplete. If that happens, repeat the fast write before playing that kit.

A backup taken from a different kit can be applied too, but the app asks about that separately — so you don't overwrite kit 12 with the contents of kit 40 by accident.

9. TD0 files

`.TD0` is the format the TD-17 uses to save a single kit to an SD card. The **TD0 file** tab lets you open such a file, look through it, change it and save a copy.

A `.TD0` file holds exactly what a kit in the module holds — instruments, levels, effects, MIDI notes. The only difference is that you are working on a file rather than on hardware, so nothing is at risk.

Things to know

- The app checks the file's checksum and also whether its values are within the ranges a real module could produce. A file that fails either check is rejected.
- Saving creates a **copy** — the original is left untouched.
- After saving, the app reads the file back to confirm it was written correctly.

Files from unknown sources

The checksum inside a TD0 file only tells you the file did not get corrupted in transit — it says nothing about who made it. Treat files downloaded from forums with the same caution as any other file from the internet.

10. Trigger settings

The **Triggers** tab shows the calibration of all ten module inputs: pad type, sensitivity, threshold, curve, scan time, mask time, retrig cancel and crosstalk.

Click **Read from module** to load them.

Read only, on purpose

The app deliberately does not let you change anything here. A badly set trigger doesn’t spoil the sound — it spoils the *playing*: the pad starts missing hits or triggering by itself. Writing will be added only once every value on this screen has been verified against the module, one by one.

Until then — if you want to change a pad’s sensitivity, do it on the module, and use this screen to check that the app sees the same value.

11. When something goes wrong

SYMPTOM	WHAT TO DO
macOS won’t open the app	System Settings → Privacy & Security → **Open Anyway** at the bottom. See chapter 2.
“The app is damaged”	The download was incomplete. Download it again from the releases page. Do not run any “repair” commands in Terminal.
The app can’t see the module	Check the TD-17 is switched on and connected by USB. Click **Refresh**. If you have several MIDI devices, select the TD-17 manually in the input and output lists.
The screen is empty, nothing loaded	Click **Full kit read**. A full read takes about three seconds.
Reading hangs or breaks off	Switch *Soft Thru* off on the module (MIDI settings). With it on, echoed notes corrupt the data frames. Also check no other program is using the TD-17 at the same time.
The module won’t confirm a write	The value may have been rejected or clamped. Check the parameter makes sense for that instrument — some V-Edit parameters exist only for certain sound families.
Clicking a pad makes no sound	Check the speaker icon in the bottom bar, the velocity slider, and whether the zone has a MIDI note assigned at all (MIDI page).
Buttons are greyed out	A MIDI operation is in progress — a read, a write, or the module’s thirty-second settling period. Wait for the status bar to go quiet.

The Log tab

It shows all communication with the module. When something misbehaves, **Copy log** puts it on the clipboard — it is the single most useful thing you can attach to a bug report.

Before pasting a log anywhere public, glance over it: it contains the names of your MIDI devices (which on many Macs are derived from the computer’s own name) and the names of your kits.

12. Where to download and how to check the file

The only official place:

`https://github.com/arkus1972/PulsoKit/releases`

Don’t download PulsoKit from anywhere else. The source is public, so anyone can copy it and publish their own build with the same description and the same icon.

Checking that a file is genuine

Every release carries a build provenance attestation from GitHub. If you have the `gh` tool installed:

```
gh attestation verify PulsoKit.zip --repo arkus1972/PulsoKit
```

This proves the file was produced by that repository, in a specific build. Next to it there is also a checksum, `PulsoKit.zip.sha256`, which you can compare using `shasum -a 256 PulsoKit.zip`.

Licence and disclaimer

PulsoKit is free and released under the MIT licence. It is written in spare time — it may contain bugs and comes with **no warranty of any kind**. You use it at your own risk; the author is not liable for lost settings or for any damage arising from its use.

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> **ONE LAST THING**
>
> The app is free and will stay that way. If it turns out to be useful you can buy the author a coffee — but nobody expects you to.